

V.I.S.O.R. Project

Systems Engineering SAT Part B: Production & Evaluation Report

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“Design is not just what it looks like and feels like. Design is how it works.”

— **Steve Jobs**

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1. Production Processes, Tools, & Safety Management

2. Technical Production & Subsystem Integration

2.1. Realisation of an integrated controlled system.

Justification of how your electronic system addresses the context, considerations and constraints of the design brief

The V.I.S.O.R. prototype directly addresses the context of school technology workshops by mitigating the “Panic Gap”—the critical delay where shock and cognitive overload prevent injured students from administering immediate first aid. By combining a multimodal Vision-Language Model (Gemini 3.7 Flash) with an active electromechanical dispensing rig, the system functions as an autonomous triage station that diagnoses injuries hands-free.

The system strictly satisfies all core constraints and considerations established in the Part A Design Brief:

1. **Latency Constraint (< 15 s):** By re-architecting the high-level logic controller in native Rust (2024 Edition) on a Raspberry Pi 5, audio downmixing executes in 2.90 ms and JSON triage parsing in $4.52\mu\text{s}$. Network transit to the Gemini API averages 3.2 s and physical servo stroke cycles require 2.4 s, yielding an end-to-end response latency of ~ 5.6 s, well beneath the 15-second threshold.
2. **Legal & Safety Constraints:** The system is hardcoded to topical wound-care consumables only (Adhesive Bandages and Isopropyl Alcohol Prep Pads); oral analgesics are strictly excluded. Electrically, 240V AC mains is isolated outside the chassis using an RCM-certified step-down adapter, powering all logic, sensors, and servos on a shared 5V DC common ground.
3. **Hygiene & Packaging Integrity:** Rather than relying on gravity drops that jam flexible wrappers, the system translates rotational servo torque into active linear horizontal thrust via an involute rack and pinion ($m \sim 1.27$ mm), contacting only the secondary outer packaging.
4. **Data Privacy:** User audio and snapshot frames are processed transiently in volatile RAM and transmitted via TLS-encrypted HTTPS; no biometric data or imagery is written to non-volatile disk storage.

Explanation of reasons how & why the final system is different to the original idea as described in the work plan

During the production and testing phase, four strategic engineering modifications were made to the original Part A concept to guarantee functional reliability and meet the submission deadline:

1. **Software Architecture Migration (Python→Rust 2024 Edition):**

- *Reason:* Initial bench testing of the Raspberry Pi 4 running Python revealed non-deterministic latency spikes (> 450 ms) caused by runtime garbage collection and dynamic typing during 16kHz PCM audio stream normalization.
- *Resolution:* The entire Pi logic pipeline was rewritten in native Rust running on a Raspberry Pi 5. This achieved zero-cost abstractions, sub-microsecond serial framing (0.87μ s), and absolute memory safety without runtime overhead.

2. **Cartridge Consolidation (3 Items→2-Item Core Triage Rig):**

- *Reason:* Analysis of school workshop injury logs indicated that minor lacerations and abrasions represent > 85% of incidents, requiring a two-stage treatment: antiseptic cleaning followed by sterile dressing. Slicing and printing three complete cartridge assemblies within the strict timeline threatened the completion of closed-loop electro-mechanical validation.
- *Resolution:* The system was consolidated to a dual-cartridge configuration (Cartridge 1: Bandages, Cartridge 2: Alcohol Pads), with the third channel (Gauze) cleanly disabled in both the Rust data structures (`DispenseItems`) and Arduino firmware (`<DISP:b,a>`). This focused engineering resources on perfecting sequential multi-item dispensing.

3. **Chassis Enclosure De-scoping (Enclosed Plywood Kiosk→Open-Frame Functional Benchtop Test-Rig):**

- *Reason:* Slicing, toolpath generation, and iterative reprints of the gear racks consumed available machine shop hours, leaving insufficient time for safe CNC routing and assembly of the 6mm pine plywood outer shell without risking wiring shorts.
- *Resolution:* The non-structural outer shell was de-scoped in favor of an Open-Frame Benchtop Functional Prototype. This mechatronic test-rig securely mounts the cartridges, dual microcontrollers, and power distribution rails, providing unobstructed diagnostic visibility into gear meshing and allowing direct oscilloscope/multimeter probing during testing.

4. **Parametric Rack Extension (95 mm → 120 mm via Fusion 360 Python API):**

- *Reason:* Physical testing revealed that while the 1.5mm gear meshing tolerance was flawless, the original 95 mm rack could not overcome the physical friction and compression of the bandage packaging against the dispensing lip, leaving items stuck halfway (85 mm actual travel).
- *Resolution:* Utilized a Fusion 360 Python script (`generate.py`) to parametrically extend the rack arm to 120.00 mm (26 teeth at 4.0 mm pitch), achieving the required stroke to guarantee 100% ejection.

Checklist Table

Critical Subsystem Implementations

1. **High-Level Command Serialization & Dispatch (`src/pi_logic/src/modules/arduino.rs`)**

Demonstrates deterministic ASCII framing with strict binary flag encoding:

1

2

```
pub fn format_dispense_command(bandage: bool, alcohol_pad: bool) → String {
```

```

3     let b = if bandage { 1 } else { 0 };
4     let a = if alcohol_pad { 1 } else { 0 };
5     format!("<DISP:{{}},{{}}>\n", b, a)
6 }
7
8 pub fn send_dispense(&mut self, bandage: bool, alcohol_pad: bool) →
   Result<(), std::io::Error> {
9     let payload = format_dispense_command(bandage, alcohol_pad);
10    debug!("Sending framed dispense command: {}", payload.trim());
11    self.send_bytes(payload.as_bytes())
12 }

```

2. Multimodal AI Medical Triage Deserializer ([src/pi_logic/src/modules/gemini.rs](#))

Demonstrates structured JSON schema parsing mapping AI diagnosis directly to hardware flags:

```

1  #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)]
2  pub struct DispenseItems {
3      pub bandage: bool,
4      pub alcohol_pad: bool,
5  }
6
7  #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)]
8  pub struct VisorAnalysis {
9      pub can_help: bool,
10     pub reasoning: String,
11     pub dispense: DispenseItems,
12     pub video_search_query: Option<String>,
13 }
14
15 // Deserialization with strict error boundaries:
16 let response_text = extract_response_text(&res)
17     .map_err(|e| std::io::Error::new(std::io::ErrorKind::InvalidData, e))?;
18 let analysis: VisorAnalysis = serde_json::from_str(response_text)?;

```

3. Low-Level Bounded Packet Parser & Actuator Loop ([arduino_control.ino](#))

Demonstrates non-blocking serial framing bounded by < and >, buffer overflow protection, and PWM power management:

```

1  void loop() {
2      while (Serial.available() > 0) {
3          char c = Serial.read();
4          if (c == '<') {
5              bufferIndex = 0;
6              serialBuffer[bufferIndex++] = c;
7          } else if (c == '>') {

```

```

8      if (bufferIndex < SERIAL_BUF_SIZE - 1) {
9          serialBuffer[bufferIndex++] = c;
10         serialBuffer[bufferIndex] = '\0';
11         commandReady = true;
12     }
13     break;
14 } else if (c != '\n' && c != '\r' && bufferIndex > 0 && bufferIndex <
SERIAL_BUF_SIZE - 1) {
15     serialBuffer[bufferIndex++] = c;
16 }
17 }
18
19 if (commandReady) {
20     processCommand(serialBuffer);
21     commandReady = false;
22     bufferIndex = 0;
23 }
24 }
25
26 void runDispenserCycle(Servo& servo, uint8_t pin) {
27     servo.attach(pin);
28     delay(50);
29     servo.write(SERVO_FORWARD);
30     delay(TIME_PUSH_MS);
31     servo.write(SERVO_STOP);
32     delay(TIME_PAUSE_MS);
33     servo.write(SERVO_REVERSE);
34     delay(TIME_RETRACT_MS);
35     servo.write(SERVO_STOP);
36     delay(50);
37     servo.detach(); // De-energizes servo coil to prevent jitter and thermal
runaway
38 }

```

4. Parametric Computational Kinematics ([cad/3d-prints/generate.py](#))

Demonstrates mathematical derivation of the 120 mm extended linear rack and mating pinion:

```

1  # Rack Parameters (Module m = tooth_pitch / pi = 4.0 / pi ~=
1.273 mm)
2  rack_height = 8.0      # 8.0 mm structural base
3  rack_length = 120.0    # Extended to 120.0 mm to guarantee full packaging
ejection
4  tooth_pitch = 4.0      # 4.0 mm circular pitch
5  tooth_height = 1.8     # 1.8 mm tooth depth
6  pinion_teeth_count = 12 # 12-tooth spur pinion

```

 Python

```
7
8 # Compute required linear tooth count along extended stroke:
9 num_teeth = int((rack_length - 15.0) / tooth_pitch) # 26 teeth
```

3. Diagnostic Test Procedures & Data Analysis

4. Project Management & Modifications

5. Evaluation & Review

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